

Requirements Traceability Matrix (RTM)									
Municipal Solid Waste Route Optimizer PES University Software Engineering Lab									
Student SRN: PES1UG24CS453 Standard IEEE 830 Traceability Specification									
Req ID	Category	Requirement Description	Priority	WBS Mapping	Architectural Component	Interface / Protocol	Test Case ID	Verification Method	Compliance Status
FR-01	Functional	Continuous ingestion of ultrasonic fill-level readings from up to 10,000 smart trash bins over cellular/NB-IoT.	Critical	WP 4.1, 4.2	Sensor Data Ingestion Service	Sensor Ingestion API (MQTT / JSON)	TC-ING-01	Integration / Benchmarking	Verified
FR-02	Functional	Dynamic threshold filtering to include only smart bins reporting fill level > 80% capacity for pickup routing.	Critical	WP 4.3	Sensor Data Ingestion Service	Telemetry Batch API (SQL / JDBC)	TC-THR-02	Automated Unit Test	Verified
FR-03	Functional	Recalculate dynamic collection routes for 50 trucks across 10,000 bins in strictly under 45 seconds.	Critical	WP 5.2, 5.3	Route Optimization Service	Route Request (gRPC / Protobuf)	TC-OPT-03	Stress / Benchmark (45s)	Verified
FR-04	Functional	Compile turn-by-turn route manifests and stream real-time task updates to truck driver mobile in-cab terminals.	High	WP 6.2	Driver Manifest Service	Dispatch API (REST / WebSocket)	TC-NAV-04	E2E Stream Verification	Verified
FR-05	Functional	Acknowledge pickup completion in real-time and reset bin fill status in spatial telemetry database.	High	WP 6.2, 4.4	Driver Manifest Service	Manifest Store API (SQL / Redis)	TC-ACK-05	Functional Integration	Verified
FR-06	Functional	Citizen portal for reporting missed garbage pickups with geo-location tagging and image upload.	High	WP 6.3	Complaint Management Service	Complaint API (REST / HTTPS)	TC-CMP-06	Functional / Regression	Verified
FR-07	Functional	Operations dashboard for sanitation supervisor displaying live GIS map, fleet status, and manual dispatch	Medium	WP 6.1	Web Portal / API Gateway	HTTPS / Web Portal API	TC-DSH-07	UI / Usability Testing	Verified
NFR-01	Performance	Route recalculation SLA <= 45 seconds for 50 trucks across 10,000 candidate bins under peak system load.	Critical	WP 5.3, 7.2	Route Optimization Service	Spatial Query API (PostGIS / SQL)	TC-PERF-01	Load Testing Benchmark	Verified
NFR-02	Scalability	Horizontal auto-scaling of telemetry ingestion tier to process >= 5,000 sensor telemetry messages per second.	High	WP 4.2, 8.2	Sensor Data Ingestion Service	MQTT Broker Clustering	TC-SCAL-02	Stress / Spike Testing	Verified
NFR-03	Reliability	Decoupled fault isolation: telemetry spikes or citizen complaint portal outages must not affect active truck	Critical	WP 3.1, 7.1	Web Portal / API Gateway	Circuit Breakers / Resiliency	TC-REL-03	Fault Injection / Chaos	Verified
NFR-04	Security	Edge TLS 1.3 encryption, JWT role-based access control (RBAC), and isolated microservice database credentials.	High	WP 3.4, 7.3	Web Portal / API Gateway	OAuth2 / JWT Token Validator	TC-SEC-04	Penetration / Security Audit	Verified
NFR-05	Availability	99.9% uptime during operational sanitation shifts (05:00 - 22:00 daily) with zero-downtime rolling deployments.	High	WP 8.2, 8.4	Infrastructure & K8s Cluster	Kubernetes Health Probes	TC-AVL-05	Uptime & Failover Audit	Verified
NFR-06	Integrity	ACID spatial transaction consistency; prevent race conditions in assigning high-priority bins across multiple	Critical	WP 3.3, 5.1	Bin & Route Database	PostGIS Spatial Transaction Engine	TC-INT-06	Concurrency Testing	Verified

	Traceability Coverage & Compliance Summary		
	Project: Municipal Solid Waste Route Optimizer		
	Metric Description	Value / Count	Traceability Status
	Total Functional Requirements (FR)	7	100% Mapped to Architecture & Tests
	Total Non-Functional Requirements (NFR)	6	100% Mapped to Architecture & Tests
	Critical Priority Requirements	6	All 6 Verified with Automated Tests
	High Priority Requirements	6	All 6 Verified with Functional Tests
	Medium Priority Requirements	1	Verified with UI/UX Acceptance Test
	Requirements with WBS Work Packages	13	100% Work Package Traceability
	Requirements with Test Cases	13	100% Test Coverage Traceability
	Overall RTM Compliance Status	100% VERIFIED	Full Bidirectional Traceability